Assignment - 03

Hotel Booking Automation Suite — Workflow Options Use Case

1. Objective

To design and build a complete automation layer in Zoho Creator for the existing "Hotel Room Booking" application, demonstrating the correct usage of all eight Workflow options available in Zoho Creator — Form Workflows, Schedules, Approvals, Payments, Blueprints, Batch Workflows, Report Workflows, and Functions — so that the booking lifecycle runs with minimal manual intervention from enquiry to post-stay follow-up.

2. Problem Statement

The Room_Booking_Form and its supporting pages and reports capture and display data well, but every downstream action currently depends on a human: sending booking confirmations, reminding guests of upcoming check-ins, approving complimentary upgrades or discounts, collecting the advance payment before confirming a room, sending feedback surveys to a large batch of checked-out guests, and recalculating the payable balance when a tariff or discount changes. This manual dependency does not scale during peak season, and it introduces delay and inconsistency.

Design a complete automation layer on top of the Room_Booking_Form in Zoho Creator, using every Workflow option Zoho Creator provides, so that routine actions happen automatically, exceptions are routed for approval, advance payments are collected and reconciled without manual follow-up, and booking status moves through a controlled, auditable process.

3. Scope of the Use Case

- Application Name: Hotel Room Booking & Front Office Management
- Base Form/Report: Room_Booking_Form, Room_Booking_Report
- Primary Users: Guests (indirect, via automated communication), Front Office, Manager, Finance Team
- Deliverable: Eight automations, one for each Zoho Creator Workflow option, layered onto the existing booking process
- Objective of the Automation Suite: Reduce manual effort, enforce a controlled approval and status process, and ensure timely communication throughout the booking lifecycle

4. Workflow Options Used and Their Mapping

The table below maps every Workflow option available in Zoho Creator to a specific automation built around the Room_Booking_Form.

Workflow Option	Workflow Name	Purpose	Trigger / Frequency
Form Workflows	On_Booking_Submit_Workflow	Sends an instant booking confirmation email to the guest and notifies the front office of the new booking	Triggered instantly on Add (form submission) of Room_Booking_Form

Workflow Option	Workflow Name	Purpose	Trigger / Frequency
Schedules	Daily_Checkin_Reminder_Schedule	Runs once a day to identify guests checking in the next day and sends a reminder email with directions and check-in instructions	Scheduled to run daily at 6:00 PM, all year round
Approvals	Complimentary_Upgrade_Approval	Routes any booking flagged for a complimentary room upgrade or special discount to the manager for approval before it is applied	Triggered on Edit whenever a front-desk staff member marks a booking for Upgrade_Requested or Special_Discount
Payments	Booking_Advance_Payment_Workflow	Generates a payment link for the advance payment amount, redirects the guest to the payment gateway, and updates Advance_Payment_Status automatically on success or failure	Triggered immediately after form submission, before the confirmation email is sent
Blueprints	Booking_Lifecycle_Blueprint	Defines and enforces the full booking lifecycle as a controlled sequence of stages with role-based transition permissions	Applied to every record in Room_Booking_Report; stages: Enquiry → Confirmed → Checked-In → Checked-Out
Batch Workflows	Bulk_Feedback_Request_Batch	Sends a post-stay feedback survey link to all guests who checked out in the last 24 hours, processed in controlled batches	Triggered automatically every night at 11:00 PM, processing records in batches of 100

Workflow Option	Workflow Name	Purpose	Trigger / Frequency
Report Workflows	OnEdit_Recalculate_Balance_Workflow	Recalculates the Total_Amount_Payable field whenever front-office staff manually edit the room tariff or discount directly from the report	Triggered on Edit of any record within the Room_Booking_Report
Functions	Generate_Invoice_Function	A reusable custom function that generates a final stay invoice PDF (room charges, taxes, and add-ons) and emails it to the guest; callable from a button, a schedule, or another workflow	Invoked automatically when a booking's Status changes to Checked-Out, and also callable manually via a 'Resend Invoice' button

5. Automation Details (Workflow-wise)

On_Booking_Submit_Workflow (Form Workflow)

- Fires the moment a new record is added to Room_Booking_Form.
- Sends a confirmation email to the guest containing their Booking ID and stay details, and notifies the front office of the new booking for room allocation.

Daily_Checkin_Reminder_Schedule (Schedule)

- Runs automatically every evening without any manual trigger, querying all records where Check_In_Date equals tomorrow's date.
- Includes directions to the property and any pre-arrival instructions configured for the hotel.

Complimentary_Upgrade_Approval (Approval)

- Only bookings explicitly flagged for a complimentary upgrade or special discount require this approval step; standard bookings bypass it entirely.
- The upgrade or discount is not applied to the guest's folio until the manager approves the request from their approval inbox.

Booking_Advance_Payment_Workflow (Payments)

- Integrates with a payment gateway connector configured in Zoho Creator to generate a secure payment link based on the configured advance payment percentage.
- On successful payment, the workflow automatically updates Advance_Payment_Status to 'Paid' and stores the transaction reference; on failure, it flags the record for front-office follow-up.

Booking_Lifecycle_Blueprint (Blueprint)

- Each stage transition (e.g., Confirmed to Checked-In) is restricted to specific roles, so a reservations agent cannot mark a guest Checked-Out without the intermediate stages being completed.
- The Blueprint view gives the manager a visual, auditable trail of exactly when and by whom each booking moved between stages.

Bulk_Feedback_Request_Batch (Batch Workflow)

- Used every night to identify all guests who checked out that day and send them a feedback survey link the following morning.
- Processes records in batches rather than all at once, preventing the workflow engine from timing out on a busy checkout day.

OnEdit_Recalculate_Balance_Workflow (Report Workflow)

- Ensures that if front-office staff manually adjust a guest's tariff or discount directly within the report grid, the Total_Amount_Payable stays accurate without requiring a full form resubmission.
- Prevents inconsistent billing that could arise from manual overrides.

Generate_Invoice_Function (Function)

- Written once as a reusable Deluge function, it can be called from multiple places: automatically when a booking's status changes to Checked-Out, manually via a 'Resend Invoice' button, or from a monthly batch that reconciles all invoices for accounting.
- Centralizing this logic in a function avoids duplicating invoice-generation code across multiple workflows.

6. Configuration & Governance Rules

- Form Workflows (On_Booking_Submit_Workflow): configured to fail gracefully and log an error if the guest's email is invalid, rather than blocking the booking.
- Schedules (Daily_Checkin_Reminder_Schedule): sends only one reminder per booking to avoid appearing repetitive.
- Approvals (Complimentary_Upgrade_Approval): configured with a single approver role (Duty Manager) and a fallback approver in case the primary approver is unavailable.
- Payments (Booking_Advance_Payment_Workflow): configured to retry the status-check callback from the payment gateway up to 3 times before flagging the transaction as failed.

- Blueprints (Booking_Lifecycle_Blueprint): each stage transition requires a mandatory comment field to be filled, ensuring an auditable reason is logged for every status change.
- Batch Workflows (Bulk_Feedback_Request_Batch): batch size configurable (default 100 records) to balance processing speed against system load.
- Report Workflows (OnEdit_Recalculate_Balance_Workflow): restricted to fire only when the Room Tariff or Discount fields specifically change, to avoid unnecessary recalculation on unrelated edits.
- Functions (Generate_Invoice_Function): includes input validation so it fails safely if called with a record that has no Guest Name or Email populated.

7. Workflow Summary (End-to-End)

- Step 1: A guest submits the Room_Booking_Form, immediately triggering the On_Booking_Submit_Workflow (Form Workflow), which sends a confirmation.
- Step 2: The Booking_Advance_Payment_Workflow (Payments) generates a payment link and updates Advance_Payment_Status once the guest completes payment.
- Step 3: If a front-desk staff member flags the booking for a complimentary upgrade, the Complimentary_Upgrade_Approval (Approvals) routes it to the manager before it can be applied.
- Step 4: The record moves through the Booking_Lifecycle_Blueprint (Blueprints) stages as staff confirm, check in, and check out the guest, each transition logged with a comment.
- Step 5: Each evening, the Daily_Checkin_Reminder_Schedule (Schedules) automatically emails guests checking in the next day.
- Step 6: If front-office staff manually adjust a tariff in the report, the OnEdit_Recalculate_Balance_Workflow (Report Workflows) keeps the payable balance accurate.
- Step 7: Every night, the Bulk_Feedback_Request_Batch (Batch Workflows) sends feedback surveys to all guests who checked out that day, processed in controlled batches.
- Step 8: The moment a guest is marked Checked-Out, a workflow calls the Generate_Invoice_Function (Functions) automatically, and staff can also trigger it manually via a button to resend an invoice.

8. Expected Learning Outcomes

- Understand the purpose and correct use case for each Workflow option in Zoho Creator.
- Learn the difference between event-driven automation (Form Workflows, Report Workflows) and time-driven automation (Schedules).
- Practice designing a controlled, auditable process using Blueprints alongside a human checkpoint using Approvals.
- Understand how Batch Workflows prevent performance issues when processing large data volumes.
- Learn how to write reusable logic once as a Function and invoke it from multiple different workflow types.
- Understand how Payments workflows integrate external gateways into a Zoho Creator process.

9. Deliverable for This Lab Exercise

Students must build all eight automations on the existing Room_Booking_Form/Report in Zoho Creator exactly as specified above, configure the listed governance rules, and submit a screenshot-based lab record showing: (a) each workflow's configuration screen in Zoho Creator's Workflow Builder, (b) the Blueprint's visual stage diagram with transition permissions, (c) a sample confirmation email and payment link triggered end-to-end, and (d) the Deluge script for the Generate_Invoice_Function along with evidence of it being called from at least two different workflow types.